Exercise 3.1 SOLUTION

Q1. Find the maximum or minimum value of the following quadratic functions by completing square.

i. $f(x)=x^2+6x+13$

Sol: $f(x)=x^2+6x+13$

$f(x)=x^2+6x+9+4$

$f(x)=(x+3)^2+4=[x-(-3)]^2+4$

Compare with

$f(x)=a(x-h)^2+k$

Here $a=1>0$

Therefore, the minimum value at $x=-3$ is 4.

نوٹ:

اگر a کی value positive تو x پر minimum اور

اگر a کی value Negative تو x پر maximum.

ii. $f(x)=x^2+4x$

Sol: Add and subtract 4 on R.H.S

$f(x)=x^2+4x+4-4$

$f(x)=(x+2)^2-4=[x-(-2)]^2+(-4)$

Compare with $a(x-h)^2+k$

Here $a=1>0$

Therefore, the minimum value at $x=-2$ is -4.

iii. $f(x)=-x^2+8x+13$

Sol: $f(x)=-x^2+8x+13$

$f(x)=-(x^2-8x)+13$

$f(x)=-(x^2-8x+16-16)+13$

$f(x)=-((x-4)^2-16)+13$

$f(x)=-(x-4)^2+16+13$

$f(x)=-(x-4)^2+29$

Compare with $f(x)=a(x-h)^2+k$

Here $a=-1<0$. Therefore, the maximum value at $x=4$ is 29

iv. $f(x)=-x^2-3x-5$

Sol: $f(x)=-x^2-3x-5$

$f(x)=-(x^2+3x)-5$

Add and subtract $\frac{9}{4}$ on R.H.S

$f(x)=-\left(x^2+3x+\frac{9}{4}-\frac{9}{4}\right)-5$

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$$f(x)=-\left[\left(x+\frac{3}{2}\right)^2-\frac{9}{4}\right]-5$$

$$f(x)=-\left(x+\frac{3}{2}\right)^2+\frac{9}{4}-5$$

$$f(x)=-\left(x-\left(-\frac{3}{2}\right)\right)^2+\frac{9-20}{4}$$

$$f(x)=-\left(x-\left(-\frac{3}{2}\right)\right)^2+\left(\frac{-11}{4}\right)$$

Compare with $a(x-h)^2+k$

Here $a=-1<0$

Therefore, the maximum value at $x=-\frac{3}{2}$ is $-\frac{11}{4}$.

v. $\boldsymbol{f(x)=3x^2+6x-13}$

Sol: $f(x)=3x^2+6x-13$

$$f(x)=3\left(x^2+2x\right)-13$$

$$f(x)=3\left(x^2+2x+1-1\right)-13$$

$$f(x)=3\left[(x+1)^2-1\right]-13$$

$$f(x)=3(x+1)^2-3-13$$

$$f(x)=3(x+1)^2-16$$

$$f(x)=3(x-(-1))^2+(-16)$$

Compare with $a(x-h)^2+k$

Here $a=3>0$. Therefore, the minimum value at $x=-1$ is -16.

vi. $\boldsymbol{f(x)=-2x^2-x+21}$

Sol: $f(x)=-2x^2-x+21$

$$f(x)=-2\left(x^2+\frac{x}{2}\right)+21$$

$$f(x)=-2\left(x^2+\frac{x}{2}+\frac{1}{16}-\frac{1}{16}\right)+21$$

$$f(x)=-2\left(\left(x+\frac{1}{4}\right)^2-\frac{1}{16}\right)+21$$

$$f(x)=-2\left(x+\frac{1}{4}\right)^2+\frac{2}{16}+21=-2\left(x+\frac{1}{4}\right)^2+\frac{1}{8}+21$$

$$f(x)=-2\left(x+\frac{1}{4}\right)^2+\frac{1+168}{8}$$

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$$f(x)=-2\left(x-\left(-\frac{1}{4}\right)\right)^2+\frac{169}{8}$$

Compare with $a(x-h)^2+k$

Here $a=-2<0$

Therefore, the maximum value at $x=-\frac{1}{4}$ is $\frac{169}{8}$.

Q2. **Find the maximum or minimum point by sketching the following quadratic functions. Also find there domai and range.**

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i. $f(x)=x^2-4x$

Sol: $f(x)=x^2-4x$

Here $a=1, b=-4, c=0$

To find y-intercept put $x=0$

$y=f(x)=(0)^2-4(0)=0$ we get $(0,0)$

To find x-intercept put $y=f(x)=0$

$0=x^2-4x$ *or* $x(x-4)=0$

$x=0$ *or* $x-4=0 \Rightarrow x=0$ *or* $x=4$

We get $(0,0)$ and $(4,0)$

To find vertex

$$h=\frac{-b}{2a}=\frac{-(-4)}{2(1)}=\frac{4}{2}=2$$

$k=(2)^2-4(2)=4-8=-4$

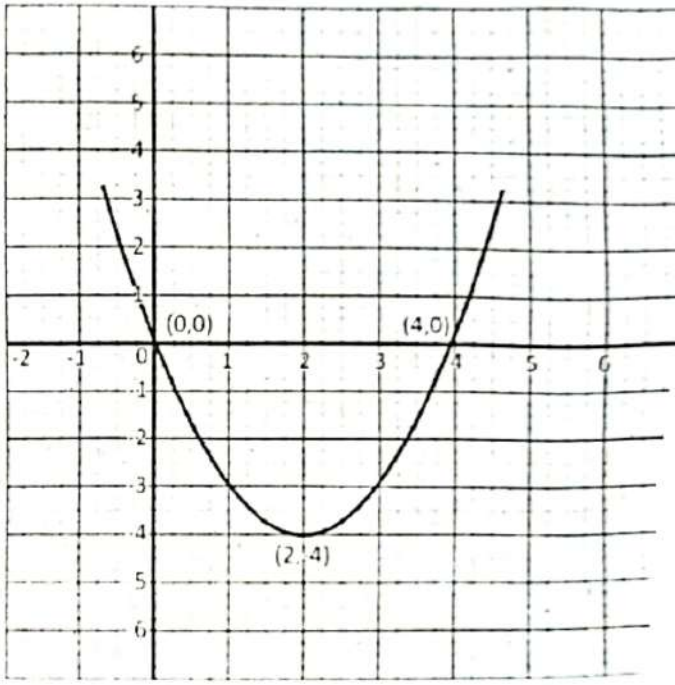
Vertex is $(2,-4)$

Domain $=(-\infty,\infty)$

Range $=[-4,\infty)$

Here $a=1>0$ so

Minimum value at $x=2$ is -4



Domain تو سب کی $(-\infty,\infty)$ ہے اگر سوال میں x کی قیمت نہ بتائی ہو اور k range کی قیمت سے ∞ تک جیسے اس سوال میں $[-4,\infty)$ تو ہے $k=-4$

ii. $f(x)=x^2-5x+6$

Sol: $f(x)=x^2-5x+6$

Here $a=1, b=-5, c=6$

To find y-intercept put $x=0$

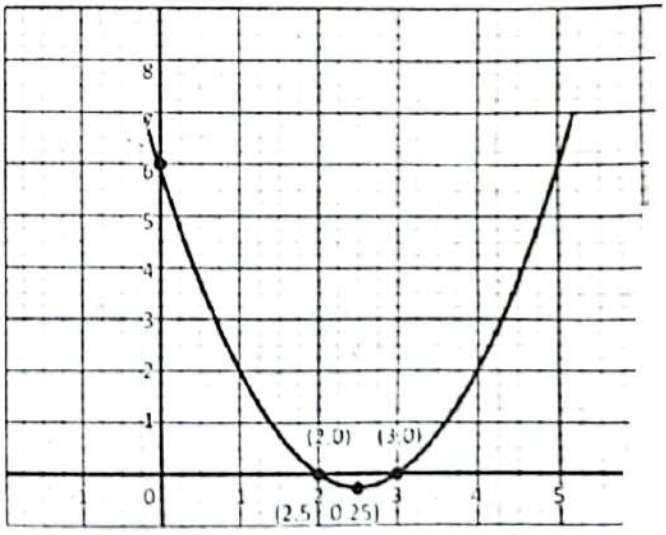
$y=f(x)=(0)^2-5(0)+6=0$ We get $(0,6)$

To find x-intercept put $y=f(x)=0$

$\Rightarrow x^2-5x+6=0$ *or* $x^2-2x-3x+6=0$

$x(x-2)-3(x-2)=0 \Rightarrow (x-3)(x-2)=0$

We get $(2,0)$ and $(3,0)$



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To find vertex

$$h=\frac{-b}{2a}=\frac{-(-5)}{2(1)}=\frac{5}{2}$$

$$k=\left(\frac{5}{2}\right)^2-5\left(\frac{5}{2}\right)+6=\frac{25}{4}-\frac{25}{2}+6=\frac{25-50+24}{4}=-\frac{1}{4}$$

Vertex is $\left(\frac{5}{2},-\frac{1}{4}\right)$

Here $a=1>0$ so

Minimum value at $x=\frac{5}{2}$ is $-\frac{1}{4}$

Domain $=(-\infty,\infty)$, Range $=[-\frac{1}{4},\infty)$

iii. $f(x)=-x^2+2x-8$

Sol: $f(x)=-x^2+2x-8$

Here $a=-1,b=2,c=-8$

For y-intercept put $x=0$

$\Rightarrow y=f(x)=-(0)^2+2(0)-8=-8$

We get $(0,-8)$

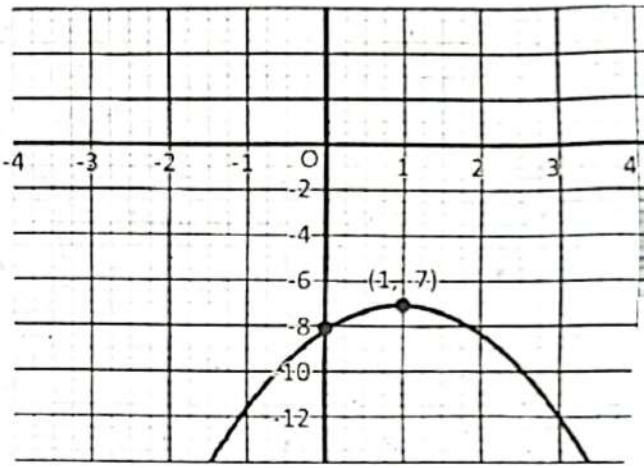
For $x=-1$ then

$y=-(-1)^2+2(-1)-8=-1-2-8=-11$

For $x=0$ then $y=-(0)^2+2(0)-8=-8$

For $x=1$ then $y=-(1)^2+2(1)-8=-1+2-8=-7$

For $x=2$ then $y=-(2)^2+2(2)-8=-4+4-8=-8$



To find vertex

$$h=\frac{-b}{2a}=\frac{-2}{2(-1)}=1$$

$$k=-(1)^2+2(1)-8=-1+2-8=-7$$

So the Vertex is $(1,-7)$

Here $a=-1<0$ so

Maximum value at $x=1$ is -7

Domain $=(-\infty,\infty)$

Range $=(-\infty,-7]$

iv. $f(x)=x^2-4x+4$

Sol: $f(x)=x^2-4x+4$

Here $a=1,b=-4,c=4$

For y-intercept put $x=0$

$\Rightarrow y=f(x)=(0)^2-4(0)+4=4$

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We get $(0,4)$

For x-intercept put $y = f(x) = 0$

$\Rightarrow x^2 - 4x + 4 = 0$ *or* $(x-2)^2 = 0 \Rightarrow x - 2 = 0$

$\Rightarrow x = 2$ we get $(2,0)$

For vertex

$$h = \frac{-b}{2a} = \frac{-(-4)}{2(1)} = \frac{4}{2} = 2$$

$$k = (2)^2 - 4(2) + 4 = 4 - 8 + 4 = 0$$

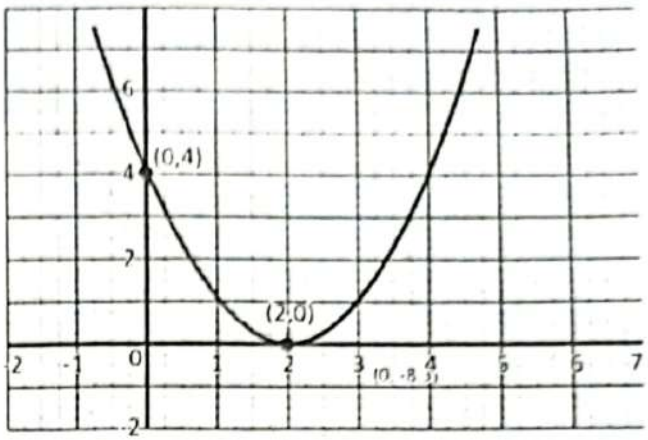
So the Vertex is $(2,0)$

Here $a = 1 > 0$ so

Therefore Minimum value at $x = 2$ is 0

Domain $= (-\infty, \infty)$

Range $= [0, \infty)$



v. $f(x) = x^2 + 2x - 8.3$

Sol: $f(x) = x^2 + 2x - 8.3$

Here $a = 1, b = 2, c = -8.3$

For y-intercept put $x = 0$

$\Rightarrow y = f(x) = 0 + 0 - 8.3 = -8.3$

We get $(0, -8.3)$

We next some other values also

Put $x = -1 \Rightarrow y = f(x) = (-1)^2 + 2(-1) - 8.3$

$= 1 - 2 - 8.3 = -9.3; (-1, -9.3)$

Put $x = 1$ $y = f(x) = (1)^2 + 2(1) - 8.3 = -5.3$

We get $(1, -5.3)$

For vertex

$$h = \frac{-b}{2a} = \frac{-2}{2(1)} = -1$$

$$k = (-1)^2 + 2(-1) - 8.3 = -9.3$$

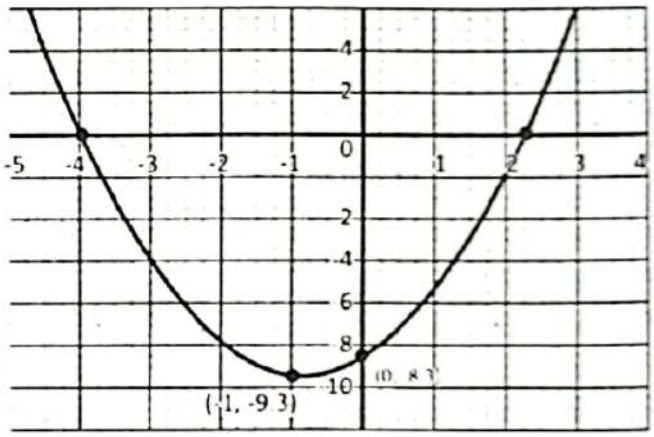
Vertex is $(-1, -9.3)$

Here $a = 1 > 0$ so

Therefore Minimum value at $x = -1$ is -9.3

Domain $= (-\infty, \infty)$

Range $= [-9.3, \infty)$



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vi. $f(x)=6-x-x^2$

Sol: $f(x)=6-x-x^2=-x^2-x+6=0$

Here $a=-1, b=-1, c=6$

For y-intercept put $x=0$

$\Rightarrow y=f(x)=6-0-(0)^2=6-0=6$

We get $(0,6)$

For x-intercept put $y=f(x)=0$

$0=6-x-x^2 \Rightarrow x^2+x-6=0$

$x^2+3x-2x-6=0$ or $x^2+x-6=0$

$(x+3)(x-2)=0 \Rightarrow x+3=0$ or $x-2=0$

$x=-3$ *or* $x=2$

We get $(-3,0)$ and $(2,0)$

For vertex

$$h=\frac{-b}{2a}=\frac{-(-1)}{2(-1)}=\frac{-1}{2}$$

$$k=6-\left(-\frac{1}{2}\right)-\left(-\frac{1}{2}\right)^2=6+\frac{1}{2}-\frac{1}{4}=\frac{24+2-1}{4}=\frac{25}{4}$$

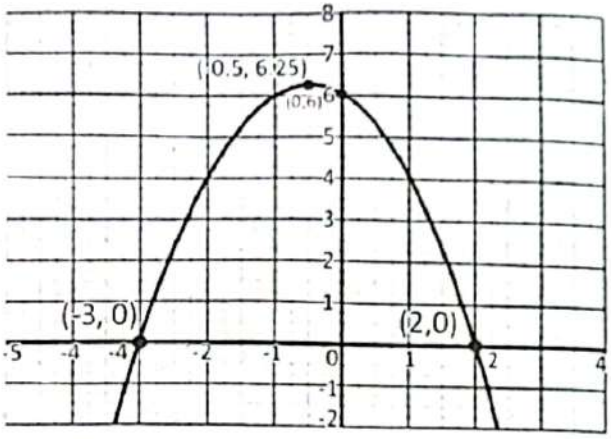
Vertex is $\left(-\frac{1}{2}, \frac{25}{4}\right)$

Here $a=-1<0$ Therefore

Maximum value at $x=-\frac{1}{2}$ is $\frac{25}{4}$

Domain $=(-\infty,\infty)$, Range $=(-\infty, \frac{25}{4}]$

نوٹ:
پہلے دیے گئے سوال کو ترتیب یعنی decending order میں لکھیں۔



Q3. Find the inverse of the following quadratic functions. Also find their domain and range.

i. $f(x)=x^2-3\ ,\ x\le 0$

Sol: $y=x^2-3$

or $x=y^2-3$ (Interchange x & y)

$\Rightarrow y^2=x+3 \Rightarrow y=\pm\sqrt{x+3}=-\sqrt{x+3}$ Because $x\le 0$

Replace y by $f^{-1}(x)$ we get $f^{-1}(x)=-\sqrt{x+3}$

Domain of $f^{-1}(x)=[-3,\infty)$

Range of $f^{-1}(x)=(-\infty,0]$

نوٹ:
پچھلے تمام سوالوں کی طرح $f(x)$ کی Domain کو Range معلوم کر کے اگر Domain کو Range میں بدل دیں اور اسی طرح Range کو Domain میں بدل دیں تو $f^{-1}(x)$ کی Domain اور Range آجائے گی

ii. $f(x)=x^2+6x+4,\quad x<-3$

Sol: $y=x^2+6x+4\ ,\ x<-3$

Let $y=x^2+6x+4$

$x=y^2+6y+4$ (Interchange x & y)

$\Rightarrow y^2+6y+4-x=0$

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$$y=\frac{-6\pm\sqrt{(6)^2-4(1)(4-x)}}{2(1)}=\frac{-6\pm\sqrt{36-16+4x}}{2}=\frac{-6\pm\sqrt{20+4x}}{2}$$

$$y=\frac{-6\pm\sqrt{4(5+x)}}{2}=\frac{-6\pm2\sqrt{5+x}}{2}=\frac{2\left(-3\pm\sqrt{5+x}\right)}{2}=-3\pm\sqrt{5+x}$$

Replace y by $f^{-1}(x)$ we get $f^{-1}(x)=-3-\sqrt{5+x}$ Because $x<-3$

Domain of $f^{-1}(x)=(-5,\infty)$ and Range of $f^{-1}(x)=(-3,\infty)$

iii. $f(x)=2x^2-8x+11\ ,\ x\geq 2$

Sol: Take $f(x)=y$ then

$y=2x^2-8x+11$

$x=2y^2-8y+11$ (Interchange x & y)

or $2y^2-8y+11-x=0$

(By quadratic formula)

$$y=\frac{-(-8)\pm\sqrt{(-8)^2-4(2)(11-x)}}{2(2)}=\frac{8\pm\sqrt{64-88+8x}}{4}=\frac{8\pm\sqrt{8x-24}}{4}$$

$$=\frac{8\pm\sqrt{4(2x-6)}}{4}=\frac{8\pm2\sqrt{2x-6}}{4}=\frac{2\left(4\pm\sqrt{2x-6}\right)}{2^2}=\frac{4\pm\sqrt{2x-6}}{2}$$

Replace y by $f^{-1}(x)$ we get

$$f^{-1}(x)=\frac{4\pm\sqrt{2x-6}}{2}=\frac{4+\sqrt{2(x-3)}}{2}$$ because $x\geq 2$

Domain of $f^{-1}(x)=[-3,\infty)$

Range of $f^{-1}(x)=[2,\infty)$

iv. $f(x)=3x^2-2x+6\ ,\ x\geq 5$

Sol: Take $f(x)=y$ then

$y=3x^2-2x+6$

$x=3y^2-2x+6$ (Interchange x & y)

or $3y^2-2y+6-x=0$

$$y=\frac{-(-2)\pm\sqrt{(-2)^2-4(3)(6-x)}}{2(3)}=\frac{2\pm\sqrt{4-72+12x}}{6}=\frac{2\pm\sqrt{12x-68}}{6}$$

$$y=\frac{2\pm\sqrt{4(3x-17)}}{6}=\frac{2\pm2\sqrt{3x-17}}{6}=\frac{1\pm\sqrt{3x-17}}{3}$$

Replace y by $f^{-1}(x)$

We get $f^{-1}(x)=\frac{1\pm\sqrt{3x-17}}{3}=\frac{1+\sqrt{3x-17}}{3}$ Because $x\geq 5$

Domain of $f^{-1}(x)=[71,\infty)$

Range of $f^{-1}(x)=[5,\infty)$

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v. $f(x)=2(x-3)^2+1\ ,\ x\geq 3$

Sol: Take $y=f(x)$ then $y=2(x-3)^2+1$

Let $x=2(y-3)^2+1$ (Interchange x & y)

$$x-1=2(y-3)^2 \Rightarrow (y-3)^2=\frac{x-1}{2}$$

$$y-3=\pm\sqrt{\frac{x-1}{2}} \Rightarrow y=3\pm\sqrt{\frac{x-1}{2}}=3+\sqrt{\frac{x-1}{2}} \qquad \text{(because } x\geq 3\text{)}$$

Replace y by $f^{-1}(x)$ we get $f^{-1}(x)=3+\sqrt{\frac{x-1}{2}}$

Domain of $f^{-1}(x)=[1,\infty)$, Range of $f^{-1}(x)=[3,\infty)$

vi. $f(x)=-3(x+4)^2-5\ ,\ x<-4$

Sol: Take $y=f(x)$ then

$y=-3(x+4)^2-5$

Let $x=-3(y+4)^2-5$ (Interchange x & y)

Or $x+5=-3(y+4)^2 \Rightarrow (y+4)^2=\frac{x+5}{-3}$

$$y+4=\pm\sqrt{\frac{x+5}{-3}} \Rightarrow y=-4-\sqrt{\frac{x+5}{-3}} \qquad \text{(because } x<-4\text{)}$$

Replace y by $f^{-1}(x)$ we get $f^{-1}(x)=-4-\sqrt{\frac{-(x+5)}{3}}$

Domain of $f^{-1}(x)=(-\infty,-5)$, Range of $f^{-1}(x)=[-\infty,-4]$

Q4. Solve the following absolute value quadratic equations and inequalities

i. $|x^2+1|=5$

Sol: $|x^2+1|=5 \Rightarrow x^2+1=\pm 5$

$x^2+1=5$ *or* $x^2+1=-5$

$x^2=5-1=4$ *or* $x^2=-5-1=-6$

$x=\pm\sqrt{4}$ *or* $x=\pm\sqrt{-6}$ (Gives imaginary values so ignore)

$x=\pm 2$

S.S=$\{-2,2\}$

نوٹ:
جب mod ختم کریں گے تو ± لگائیں گے ایک equation کے ساتھ جمع اور دوسری کے ساتھ منفی

ii. $|x^2+5x+4|=0$

Sol: $|x^2+5x+4|=0 \Rightarrow x^2+5x+4=\pm 0$

$\Rightarrow x^2+5x+4=0 \Rightarrow x^2+x+4x+4=0$

$x(x+1)+4(x+1)=0 \Rightarrow (x+4)(x+1)=0$

$x+1=0$ *or* $x+4=0 \Rightarrow x=-1$ *or* $x=-4$

S.S=$\{-1,-4\}$

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iii. $|x^2-6x+8|=4$

Sol: $|x^2-6x+8|=4 \Rightarrow x^2-6x+8=\pm 4$

$x^2-6x+8=4$ *or* $x^2-6x+8=-4$

$x^2-6x+4=0$ *or* $x^2-6x+12=0$

$$x=\frac{-(-6)\pm\sqrt{(-6)^2-4(1)(4)}}{2(1)} \text{ or } x=\frac{-(-6)\pm\sqrt{(-6)^2-4(1)(12)}}{2(1)}$$

$$x=\frac{6\pm\sqrt{36-16}}{2}=\frac{6\pm\sqrt{20}}{2}=\frac{6\pm\sqrt{4\times 5}}{2} \text{ or } \frac{6\pm\sqrt{36-48}}{2}=\frac{6\pm\sqrt{-12}}{2}$$ (Imaginary ignore)

$$x=\frac{6\pm 2\sqrt{5}}{2}=3\pm\sqrt{5}$$

S.S $=\{3-\sqrt{5},3+\sqrt{5}\}$

iv. $|3x^2-7x+2|=x^2-x+1$

Sol: $|3x^2-7x+2|=x^2-x+1$

$3x^2-7x+2=\pm(x^2-x+1)$

$3x^2-7x+2=x^2-x+1$ *or* $3x^2-7x+2=-(x^2-x+1)$

$3x^2-7x+2-x^2+x-1=$ *or* $3x^2-7x+2+x^2-x+1=0$

$2x^2-6x+1=0$ *or* $4x^2-8x+3=0$

$$x=\frac{-(-6)\pm\sqrt{(-6)^2-4(2)(1)}}{2(2)} \quad \text{or} \quad x=\frac{-(-8)\pm\sqrt{(-8)^2-4(4)(3)}}{2(4)}$$

$$x=\frac{6\pm\sqrt{36-8}}{4}=\frac{6\pm\sqrt{28}}{4} \quad \text{or} \quad x=\frac{8\pm\sqrt{64-48}}{8}=\frac{8\pm\sqrt{16}}{8}=\frac{8\pm 4}{8}$$

$$x=\frac{6\pm 2\sqrt{7}}{4}=\frac{3\pm\sqrt{7}}{2} \quad \text{or} \quad x=\frac{8+4}{8}=\text{and } x=\frac{8-4}{8}$$

$$x=\frac{3-\sqrt{7}}{2} \text{ and } x=\frac{3+\sqrt{7}}{2} \quad \text{or} \quad x=\frac{12}{8} \text{ and } x=\frac{4}{8}$$

$$\text{S.S}=\left\{\frac{1}{2},\frac{3}{2},\frac{3-\sqrt{7}}{2},\frac{3+\sqrt{7}}{2}\right\} \quad \text{or} \quad x=\frac{3}{2} \text{ and } x=\frac{1}{2}$$

v. $|x^2-4|<5$

Sol: $|x^2-4|<5 \Rightarrow -5<x^2-4<5$

$-5<x^2-4$ *and* $x^2-4<5$ *and* $x^2<9$

$0<x^2+1$ *and* $x<\pm 3 \Rightarrow -3<x<3$

$x^2+1>0 \Rightarrow x^2>-1$

$x>\pm\sqrt{-1}$ (imaginary ignore)

S.S $=\{(-3,3)\}$

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vi. $|x^2-3x+2|>4$

Sol: $|x^2-3x+2|>4$

$\Rightarrow x^2-3x+2<-4$ *and* $x^2-3x+2>4$

$x^2-3x+6<0$ *and* $x^2-3x-2>0$

Take $x^2-3x+6=0$ Take $x^2-3x-2=0$

$x=\dfrac{-(-3)\pm\sqrt{(-3)^2-4(1)(6)}}{2(1)}$ *and* $x=\dfrac{-(-3)\pm\sqrt{(-3)^2-4(1)(-2)}}{2(1)}$

$x=\dfrac{3\pm\sqrt{9-24}}{2}$ *and* $x=\dfrac{3\pm\sqrt{9+8}}{2}$

$x=\dfrac{3\pm\sqrt{-15}}{2}$ *and* $x=\dfrac{3\pm\sqrt{17}}{2}$

$x=\dfrac{3-\sqrt{17}}{2}$ *and* $x=\dfrac{3+\sqrt{17}}{2}$

Because $x^2-3x+6<0$ Because $x^2-3x+2>0$

$\Rightarrow x<\dfrac{3\pm\sqrt{-15}}{2}$ (Imaginary ignore) $\Rightarrow x>\dfrac{3-\sqrt{17}}{2}$ *and* $x>\dfrac{3+\sqrt{17}}{2}$

$S.S=\left\{\left(-\infty,\dfrac{3-\sqrt{17}}{2}\right)\cup\left(\dfrac{3+\sqrt{17}}{2},\infty\right)\right\}$

vii. $|x^2-5x+6|\le x+2$

Sol: $|x^2-5x+6|\le x+2$

$|x^2-5x+6|=x+2$ *and* $|x^2-5x+6|<x+2$

$x^2-5x+6=\pm(x+2)$ *and* $-(x+2)<x^2-5x+6<x+2$

$x^2-5x+6=x+2$ *or* $x^2-5x+6=-(x+2)$ *and* $-x+2<x^2-5x+6$ *and* $<x^2-5x+6<x+2$

$x^2-5x+6\cdot x-2=0$ *or* $x^2-5x+6+x+2=0$ *and* $0<x^2-5x+6+x-2$ *and* $<x^2-5x+6-x-2<0$

$x^2-6x+4=0---I$ *or* $x^2-4x+8=0---II$ *and* $0<x^2-4x+4>0---III$ *and* $<x^2-6x+4<0---IV$

$\Rightarrow x^2-6x+4\le 0$ (combine I and IV)

Take $x^2-6x+4=0$

$x=\dfrac{-(-6)\pm\sqrt{(-6)^2-4(1)(4)}}{2(1)}$ *or* $x=\dfrac{-(-4)\pm\sqrt{(-4)^2-4(1)(8)}}{2(1)}>0$

$x=\dfrac{6\pm\sqrt{36-16}}{2}$ *or* $x=\dfrac{4\pm\sqrt{16-32}}{2}$

$x=\dfrac{6\pm\sqrt{20}}{2}$ *or* $x=\dfrac{4\pm\sqrt{-16}}{2}$

$x=\dfrac{6\pm\sqrt{4\times5}}{2}$ *or* *imaginary ignore*

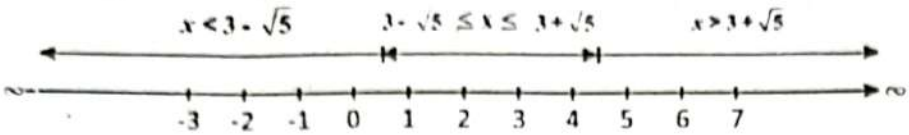
$(x-2)^2>0$

$x-2>0$

$x>2$

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$$x=\frac{6\pm2\sqrt{5}}{2}=\frac{6\pm2\sqrt{5}}{2}=3\pm\sqrt{5}=3\pm\sqrt{5}$$



Test $x=0,\ x=1,\ x=6,\ in\ x^2-6x+4\le0$

At $x=0;\ (0)^2-6(0)+4\le0\Rightarrow4\le0$ False

At $x=1;\ (1)^2-6(1)+4\le0\Rightarrow-2<0$ True

At $x=6;\ (6)^2-6(6)+4\le0\Rightarrow4\le0$ False

viii. $\left|2x^2-3x-5\right|<4$

Sol: $\left|2x^2-3x-5\right|<4$

$-4<2x^2-3x-5<4$

$-4<2x^2-3x-5 \quad and \quad 2x^2-3x-5<4$

$2x^2-3x-5+4>0 \quad and \quad 2x^2-3x-9<0$

$2x^2-3x-1>0$

Take $2x^2-3x-1=0$

$$x=\frac{-(-3)\pm\sqrt{(-3)^2-4(2)(-1)}}{2(2)}$$

$$x=\frac{3\pm\sqrt{9+8}}{4}$$

$$x=\frac{3\pm\sqrt{17}}{4}$$

$$x=\frac{3-\sqrt{17}}{4} \quad and \quad x=\frac{3+\sqrt{17}}{4}$$

Because $2x^2-3x-1>0$

$$\Rightarrow x>\frac{\sqrt{17}-3}{4} \quad and \quad x>\frac{-\sqrt{17}-3}{4}$$

$$S.S=\left\{\left(-\frac{3}{2},\frac{-\sqrt{17}+3}{4}\right)\cup\left(\frac{\sqrt{17}+3}{4},3\right)\right\}$$

Take $2x^2-3x-9=0$

$$and \quad x=\frac{-(-3)\pm\sqrt{(-3)^2-4(2)(-9)}}{2(2)}$$

$$and \quad x=\frac{3\pm\sqrt{9+72}}{4}$$

$$and \quad x=\frac{3\pm\sqrt{81}}{4}$$

$$and \quad x=\frac{3\pm9}{4}$$

$$x=\frac{12}{4} \quad and \quad x=\frac{3-9}{4}$$

$$x=3 \quad and \quad x=\frac{-6}{4}$$

$$x=3 \quad and \quad x=\frac{-3}{2}$$

Because $2x^2-3x-9<0$

$$\Rightarrow x<3 \quad and \quad x<\frac{-3}{2}$$

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